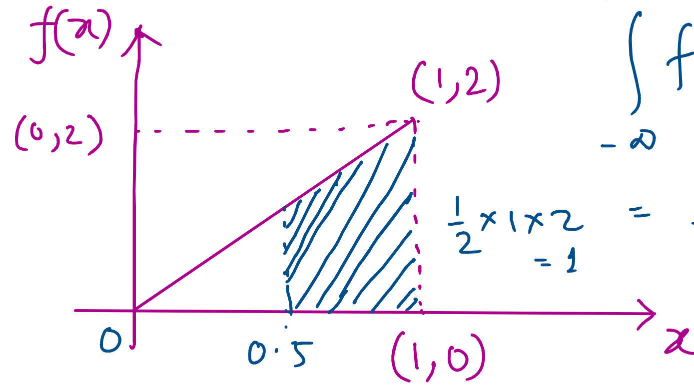
$$f(x) = \begin{cases} 2x & ; \quad 0 \le x \le 1 \\ 0 & ; \quad \text{otherwise} \end{cases}$$



$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\frac{1}{2}\times 1 \times 2 = \int_0^1 f(x)\,dx$$

$$= 1$$

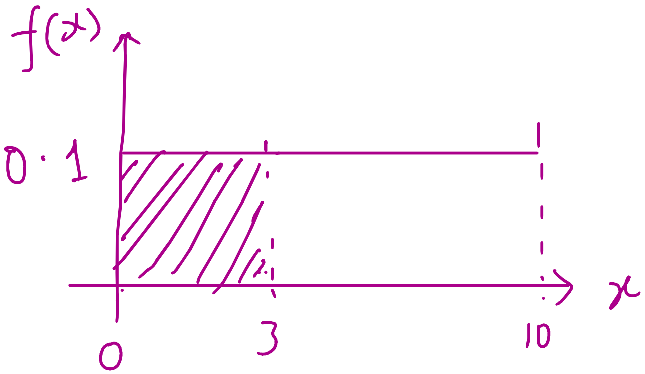
$P(X > 0.5)$ = area of the shaded region

$$P(X > 0.5) = \int_{0.5}^{1} (2x)\,dx = \cancel{2}\,\frac{x^2}{\cancel{2}} = x^2\Big]_{0.5}^{1} = 1 - (0.5)^2$$

$$= 1 - 0.25 = 0.75$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f(x)\,dx = \int_0^1 x \cdot (2x)\,dx = 2\int_0^1 x^2\,dx$$

$$= 2 \cdot \left[\frac{x^3}{3}\right]_0^1 = 2 \cdot \frac{1}{3} = \frac{2}{3}$$



discrete: $\mathbb{E}[X] = \sum_{x} x \, p_X(x)$

cont: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \, f(x) \, dx$

discrete: $\mathrm{Var}(X) = \sum_{x} (x-\mu)^2 \, p_X(x)$ where, $\mu = \mathbb{E}[X]$

cont: $\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) \, dx$

$\sigma_X = \sqrt{\mathrm{Var}(X)}$

$\mathrm{Var}(X)$ $\qquad\qquad \mu = \mathbb{E}[X] = \frac{2}{3}$

$$f(x) = \begin{cases} 2x & ; 0 \le x \le 1 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\underbrace{\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}$$

Valid for both discrete & continuous distribution.

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_0^1 x^2 \cdot (2x)\, dx = 2\int_0^1 x^3\, dx$$

$$= 2 \cdot \left[\frac{x^4}{4}\right]_0^1 = \frac{2}{4} = 0.5$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$= \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9-8}{18} = \frac{1}{18}$$